BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 187 · Linear search: trace and code

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What property must both inputs satisfy before a standard merge step?

Q2. What does the key represent in insertion sort?

Q3. What is the last index of a 5-element C# array?

Q4. A loop checks a[0], a[1], a[2] then returns. How many comparisons were made?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

target — Value the search seeks.

index — Zero-based position of an array element.

sentinel — Special result such as -1 meaning not found.

early return — Leave the function as soon as a match is found.

O(n) — Search may inspect every item as input grows.

New learning

Begin at index zero — Linear search examines one value at a time, in order.

Stop at the first match — When a[i] equals target, return i immediately.

No match — After scanning every element, return -1; no ordering assumption is needed.

Worked example

Find 16 in [6,11,16,21]. Give the index and number of comparisons.

1. Predict first — Write visited indexes before identifying the match.

2. Model solution — Visit 0 (6), 1 (11), 2 (16). Return index 2 after three comparisons.

We Do · Practise with guidance

Find 18 in [2,7,13,18,25].

Predict first

Which indexes will be visited?

Model solution

0, 1, 2, 3. Return index 3 after four comparisons.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Search for 7 in [4,7,9,12]. Give index and comparison count.

Question 2

Search for 5 in [3,8,11,16]. Give return value and count.

Question 3

Modify the function to count comparisons without returning early; explain the output for a 5-element array.

Exit Ticket · Answer before checking

Linear search requires sorted input.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Each input run must already be sorted.

Q2. The value being inserted into the sorted prefix.

Q3. 4. Indexes are 0 through 4.

Q4. 3 comparisons.

You Do pages · Independent questions

Q1. Index 1, after two comparisons.

Q2. Return -1 after all four comparisons.

Q3. Track int comparisons=0 and increment each iteration; if searching all elements, comparisons=5. To report both location and count, return a tuple (index, comparisons) or use an out parameter.

Exit Ticket · Model answer

False. It works on unsorted arrays.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: